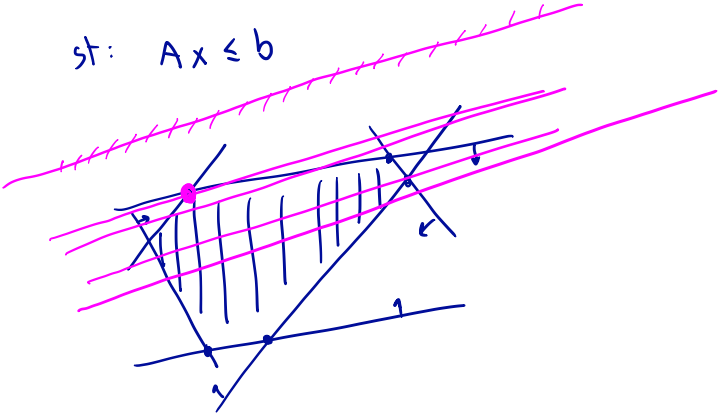
Solving LPs

Recall Geometry of LPs

min: $c^T x$

st: $Ax \le b$



Solving LPs

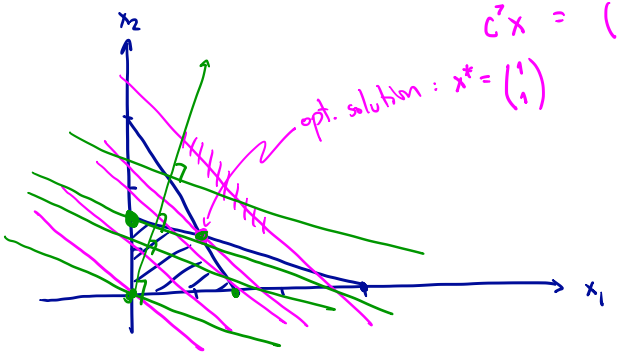
An Example. Solving by Drawing Level Sets

$$\max: \; x_1 + x_2$$

$$\text{st}: \; x_1 + 2x_2 \leq 3$$

$$2x_1 + x_2 \leq 3$$

$$x_1, x_2 \geq 0$$



$$c^T x = \begin{pmatrix} 1 \\ 1 \end{pmatrix}^T \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$c = \begin{pmatrix} 1 \\ 3 \end{pmatrix} : \; \max: \; x_1 + 3x_2$$

Solving large scale LPs

Algorithms:

Simplex Algorithm

Interior Point Methods

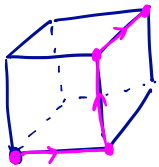
Software: Gurobi
CPLEX
others...

Solving large scale LPs

Simplex Algorithm

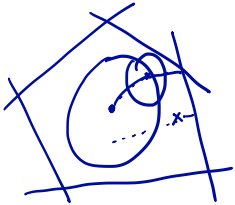
Key idea: opt. solution of LP always at an extreme point.

Simplex:



Solving large scale LPs

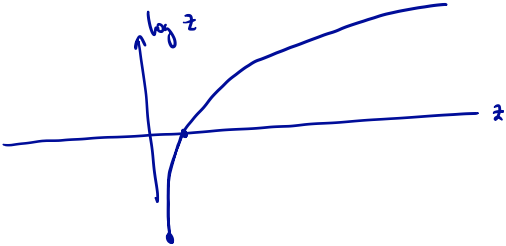
Interior Point Methods



Key idea:

$$\min: \ c^T x$$

$$\text{st}: \ a_i^T x \le b_i, \quad i = 1, \dots, m$$

$$\min: \ t \cdot c^T x - \boxed{\sum_{i=1}^{m} \log(b_i - a_i^T x)}$$

use Newton's Method.